

- Expected $pCO_2 = 15 + 9 \rightarrow 24$ mmHg
- Compensated
- Anion gap = 135 (116+9) = 10 meq/L
- NAGMA (Normal anion gap metabolic acidosis)

CASE - 8

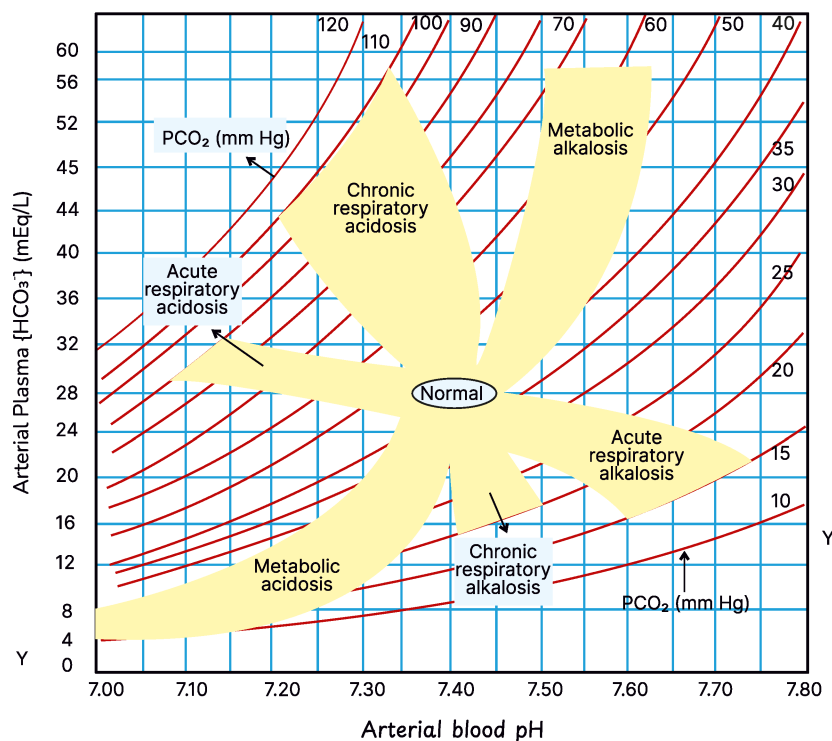
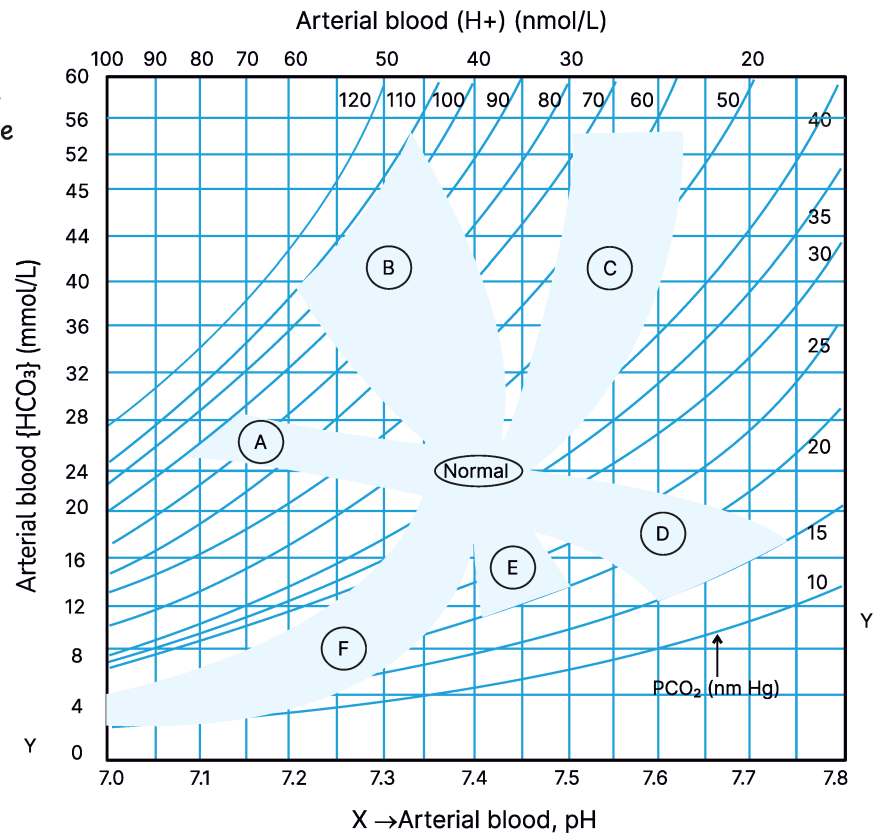
In the given acid-base nomogram diagram, areas marked A and D indicate which of the govern options?

- Chronic respiratory acidosis, acute metabolic alkalosis
- Acute metabolic alkalosis, acute respiratory acidosis
- Acute respiratory acidosis, acute respiratory alkalosis
- Acute respiratory acidosis, acute metabolic alkalosis

Ans: (c)

INFERENCE

- Image - Davenport diagram
- Mostly used by the paramedical professionals



CASE - 9

A 40-year-old man presented with nausea, dizziness, and hyperventilation. His arterial blood gas analysis and S. Creatinine revealed metabolic acidosis with a normal anion gap. State the possible reason for this condition?

- a. Lactic acidosis
- b. Chronic renal failure
- c. Renal tubular acidosis
- d. Methanol toxicity

Ans: (c)

INFERENCE

- Patient with nausea, dizziness, hyperventilation
- Metabolic acidosis with normal anion gap
- Causes of NAGMA being asked
- Lactic acidosis
- Chronic renal failure
- Methanol toxicity
- NAGMA (Normal anion gap metabolic acidosis)
 - DR. FUSE (mnemonic)
 - Diarrhea
 - Renal tubular acidosis
 - Fistula
 - Ureterosigmoidostomy

CASE - 10

Which of the following is not associated with Type A lactic acidosis?

- a. Cardiogenic shock due to COVID-19 myocarditis
- b. Acute pyelonephritis with septic shock
- c. Cyanide toxicity
- d. Metformin

Ans: (d)

INFERENCE

- Type A lactic acidosis → d/t ↓ perfusion to tissues (cardiogenic shock)
- Septic shock → ↓ perfusion → type A lactic acidosis
- Cyanide toxicity → ↓ O₂ → anaerobe respiration → lactic acid accumulation → type A lactic acidosis
- Metformin can cause Type B lactic acidosis

CASE - 11

Hepatitis B patient with liver cirrhosis develops massive ascites and SOB. He is on spironolactone plus Lasix. ABG report shows:

pH=7.55

pCO₂ = 38 mm Hg

HCO₃⁻ = 33 meq/L

Na⁺ = 140 meq/L

Chloride = 91 meq/L

AG = 16 meq/L

- Metabolic alkalosis with respiratory acidosis
- Metabolic alkalosis with respiratory alkalosis
- HAGMA
- NAGMA

Ans: (b)

INFERENCE

- To resolve SOB → ascites should be reduced → Lasix and Spironolactone
- pH - ↑, Normal pCO₂, HCO₃⁻ ↑
- Metabolic alkalosis
- Expected pCO₂ = 33 + 15 → 48 mmHg
- Actual pCO₂ = 38 mmHg
- Compensation process is not working → mixed disorder (metabolic alkalosis with respiratory alkalosis)
- pCO₂ is not elevated d/t SOB

Metabolic acidosis with respiratory acidosis	- Pulmonary edema d/t CHF ↓ perfusion → Metabolic acidosis - Severe pulmonary edema → respiratory acidosis
Metabolic alkalosis with respiratory alkalosis	- Cirrhosis pt on diuresis - Ascites → 2° aldosteronism → Metabolic alkalosis - Massive ascites → SOB → CO₂ washout respiratory alkalosis
Metabolic acidosis with metabolic alkalosis	CKD with vomiting/ diuretics
HAGMA with NAGMA	Toluene toxicity, Diarrhea + lactic acidosis



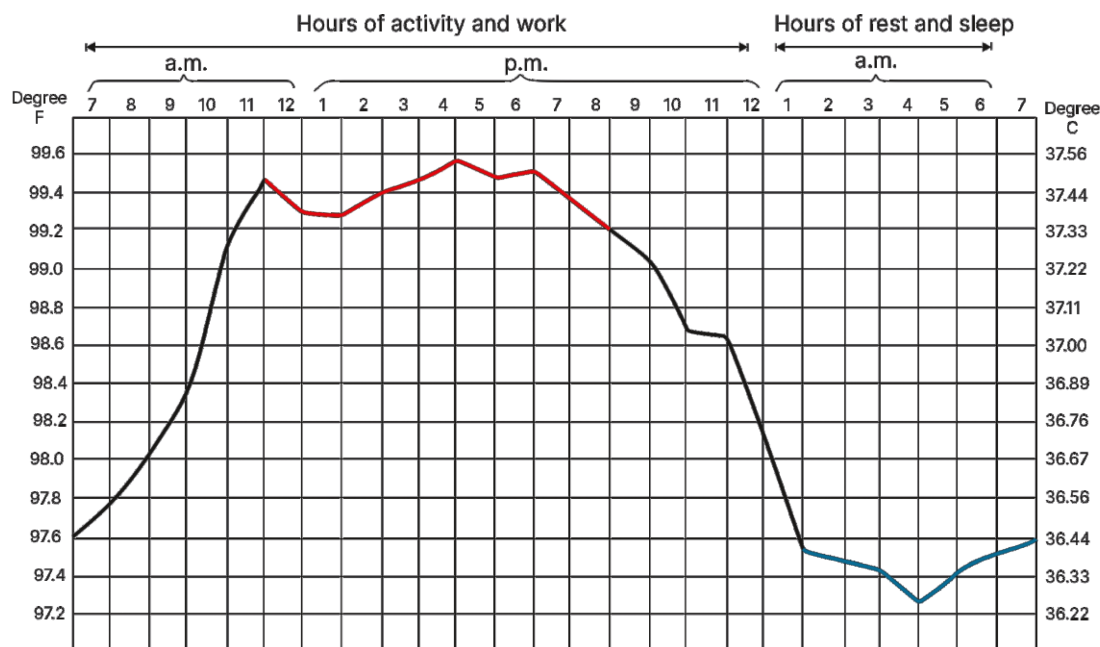
BR. FEVER OF UNKNOWN ORIGIN



DIURNAL VARIATION OF TEMPERATURE

00:00:38

- Maximum body temperature is seen between 2PM - 6PM (d/t high muscular activity)
- Least body temperature is seen between 2AM - 6AM (d/t relaxed body muscles)
- Normal diurnal variation in body temperature - $< 0.9^{\circ}\text{F}$



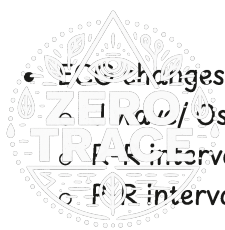
CUT OFF VALUES

00:01:20

FEVER	$>99.9^{\circ}\text{F} / 37.7^{\circ}\text{C}$	Oral temperature
FEVER OF UNKNOWN ORIGIN	$>101^{\circ}\text{F}$ for duration of >3 weeks on two separate occasions	
HYPOTHERMIA	$<95^{\circ}\text{F} / 35^{\circ}\text{C}$	Core temperature
HYPERPYREXIA	$>106.7^{\circ}\text{F} / 41.5^{\circ}\text{C}$	
HEAT STROKE	$>40.5^{\circ}\text{C}$ + CNS malfunction $\pm$ Sweating	

CORE TEMPERATURE MONITORING SITES

- **Ideal site:** Pulmonary artery by Swan Ganz catheter
- **Preferred site:** Lower esophagus > Rectum
 - Disadvantage of rectal temp. - Probe may record the temperature of frozen feces present there
- **Grades of hypothermia**



- ECG changes in hypothermia
 - P-R-T/ Osborn wave
 - P-R interval $\uparrow$
 - Q-R interval $\uparrow$

MILD	32 - 35° C
MODERATE	28 - 32° C
SEVERE	< 28° C

RECTAL THERMOMETER

- Indicated in children and in patients who will not or cannot cooperate fully with oral thermometers
- Rectal thermometer is a lubricated thermometer inserted 4 to 5cm into the anal canal at an angle 20° with the patient lying prone
 - Dwell time: 3 minutes



Rectal thermometer



Adult rectal thermometer

- Rectal temperature - 0.7° F > Oral temperature
- Axillary temperature - 1° F < Oral temperature

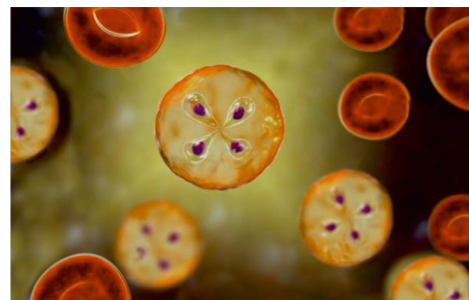
FAGET'S SIGN/ RELATIVE BRADYCARDIA

00:09:27

- For every 1° F rise of temperature above 100° F $\rightarrow$ PR/ HR $\uparrow$ by 10 beats/min

CAUSES OF RELATIVE BRADYCARDIA

- Bacterial infections: (Mnemonic: TBM CLR)
 - Typhoid, Tularaemia or Typhus Scrub
 - Brucellosis or Babesia
 - Mycoplasma pneumonia or Malaria
 - Coxiella burnetii (Q fever), Corynebacterium diphtheria
 - Legionella pneumophila, Leptospira
 - Rickettsia



Maltese cross appearance

TULAREMIA	• Common in hunters • Caused by bite of mites (Living on the bodies of rabbits) • Eschar formation and axillary lymphadenopathy
SCRUB TYPHUS	• Common in farmers (Kashmir and Tamil Nadu) • Caused by bite of baby mites (Chiggers)
BRUCELLOSIS	• Aka Undulant fever or Malta fever • Special culture media: Castaneda media (Biphasic media)
BABESIA MICROTI	• On peripheral smear: Maltese cross appearance



COXIELLA BURNETII

- Transmitted by contact with animals like Sheep, Goats
- Cause Infective endocarditis

LEGIONELLA PNEUMOPHILA

- Cause pneumonia
- A/w SIADH

LEPTOSPIROSIS

- Transmitted by contamination of drinking water by rat urine
- Cause hepatorenal syndrome (Lepto ictero haemorrhagica)

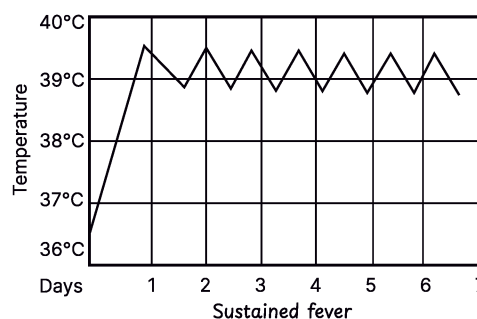
- Viral infections
 - Dengue fever: Biphasic fever
 - Yellow fever
- Drugs fever
 - Allopurinol
 - Sulfa drugs
 - Diuretics (Furosemide, Torsemide)
- CNS lesions
 - Pontine stroke
 - Pinpoint pupil
 - Cause damage to parabrachial nucleus that has negative control on hypothalamic thermoregulatory set point → High grade fever
- Lymphoreticular malignancy
 - Lymphoma

PATTERNS OF FEVER

00:17:53

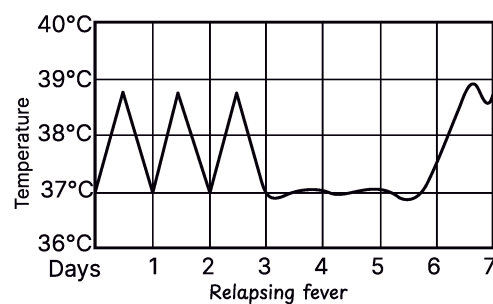
CONTINUOUS/SUSTAINED FEVER

- Diagonal variation of temperature
- Fever never touches baseline
- Fluctuation: $< 1.1^{\circ}\text{C}$



RELAPSING FEVER

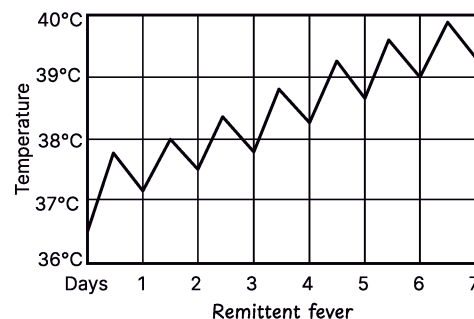
- Recurrent episodes of fever with afebrile periods in between
- Example: Lyme's disease





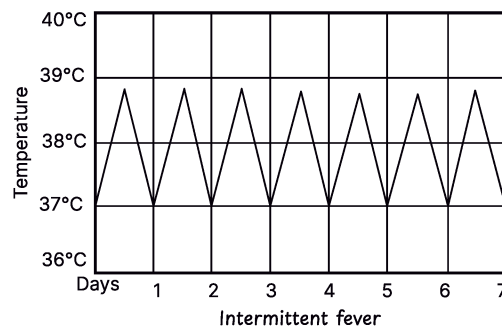
REMITTENT FEVER

- Fever never touches baseline
- Fluctuation: $>1.1^{\circ}\text{C}$



INTERMITTENT FEVER

- Fever touches the baseline
- Recurrent episodes of fever with **no afebrile periods** in between.

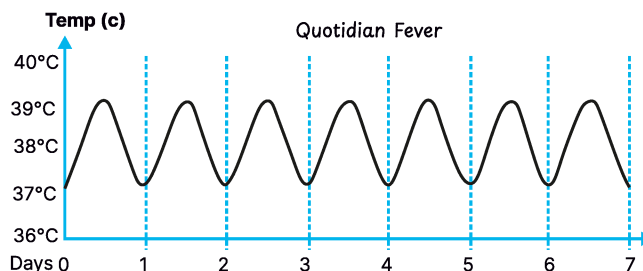


PATTERNS OF FEVER IN MALARIA, TYPHOID, ACUTE TONSILLITIS

00:21:06

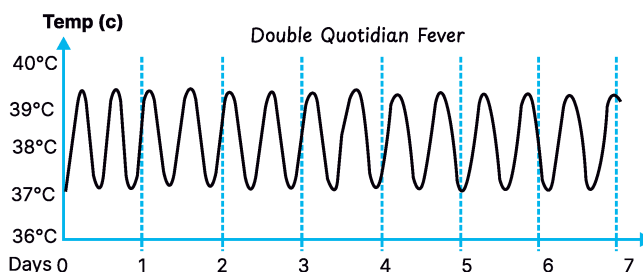
QUOTIDIAN FEVER

- Fever recurs every 24 hrs
- Seen in *P. Falciparum* (Affects all stages of RBCs) and *P. Knowlesi*



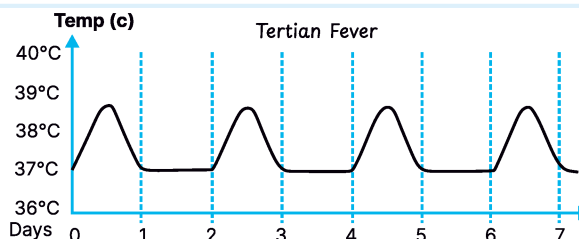
DOUBLE QUOTIDIAN FEVER

- Fever spikes twice per day
- Seen in Adult-onset Still's disease (AOSD)



TERTIAN FEVER

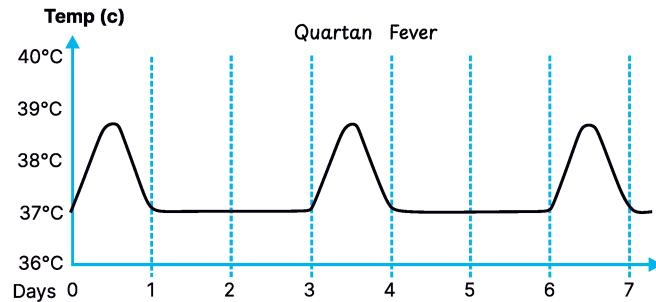
- Fever spikes every 48 hours
- Seen in *P. Vivax*, *P. Ovale*





QUARTAN FEVER

- Fever spikes every 72 hours
- Seen in *P. Malariae* (Affects mature RBCs)



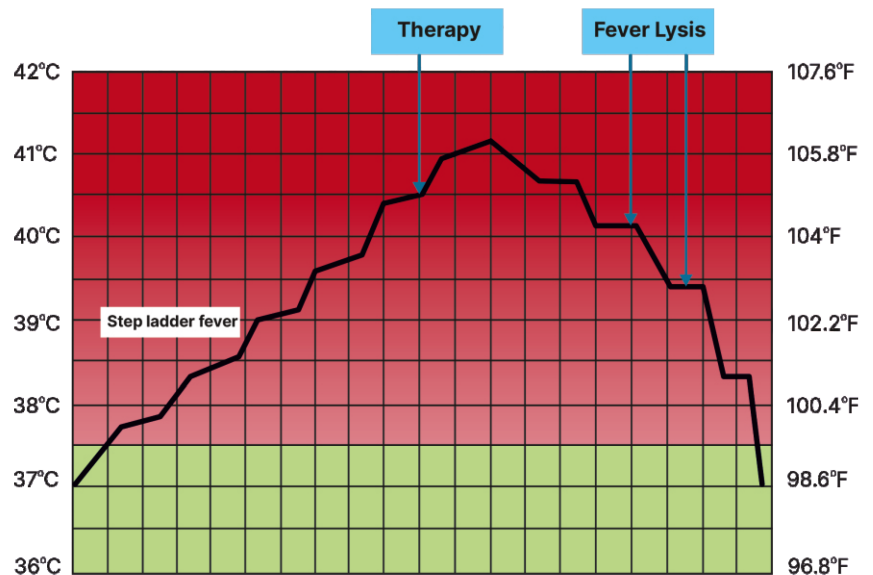
FEVER RESOLUTION WITH LYSIS

- Seen in Typhoid post antibiotic (Ceftriaxone) use

DAY		1		2		3		4		5		6		7	
TIME		AM		AM		AM		AM		AM		AM		AM	
F	°C														
105	43.6														
104	40.0														
130	39.4														
102	38.9														
101	38.3														
100	37.8														
99	37.2														

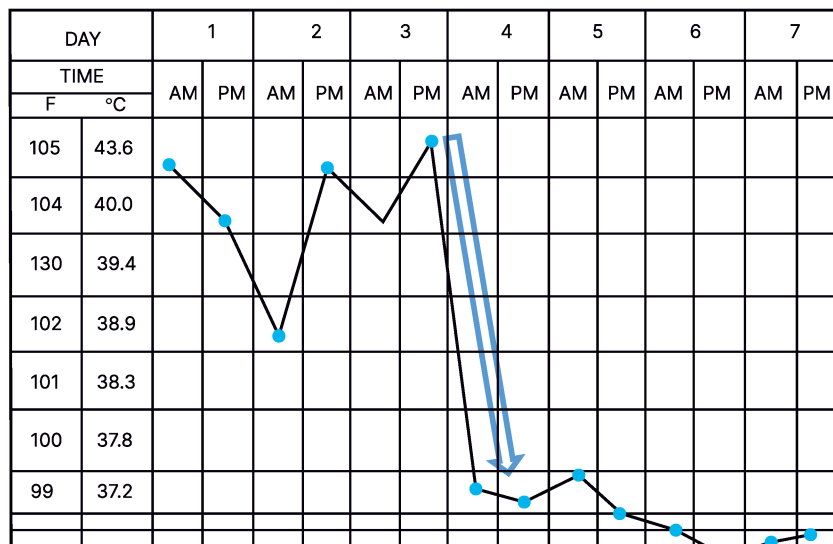
STEP LADDER PATTERN OF FEVER

- Seen in Typhoid





- Steep fall in temperature
- Seen in Acute tonsillitis, Lobar pneumonia



- Katayama fever: Seen with Schistosomiasis
- Q fever: Seen with Coxiella burnetii
 - Transmitted by contact with animals like Sheep, Goats
 - Cause Bacterial endocarditis
- Parrot fever: Seen with Chlamydia Psittaci

PYREXIA OF UNKNOWN ORIGIN

00:26:06

- **Essential components for diagnosis**
 - Fever: $>101^{\circ}\text{F}$ on 2 separate occasions
 - Duration: > 3 weeks
 - Absence of immunocompromised state
 - Uncertain diagnosis: After a thorough
 - Clinical history-taking: History with respect to travel, pet exposure, exposure to wild animals shooting etc
 - Physical examination: Evaluation of lymph nodes, enlarged liver or spleen
 - Obligatory investigations

OBLIGATORY INVESTIGATIONS

00:28:27

- Hb, TLC, DLC, Platelet count, and ESR ($\uparrow$ in Autoimmune disease)
- Acute phase reactants: CRP, LDH, and Ferritin
 - Elevated in Inflammatory illness (disseminated TB in India)
- RFT and Serum electrolytes
- LFT: SGOT/SGPT/SAP
- Serum electrophoresis - To rule out Multiple myeloma
- Autoimmune panel: Creatinine Kinase, RA factor and ANA
- Urine microscopy and culture (single culture is sufficient)
- Blood culture - **3 cultures should be negative**
- Chest X-ray/USG/TST (Tuberculin skin test)/IGRA (Interferon Gamma Release Assay)

STEPS IN WORKUP OF FUO AFTER OBLIGATORY INVESTIGATIONS

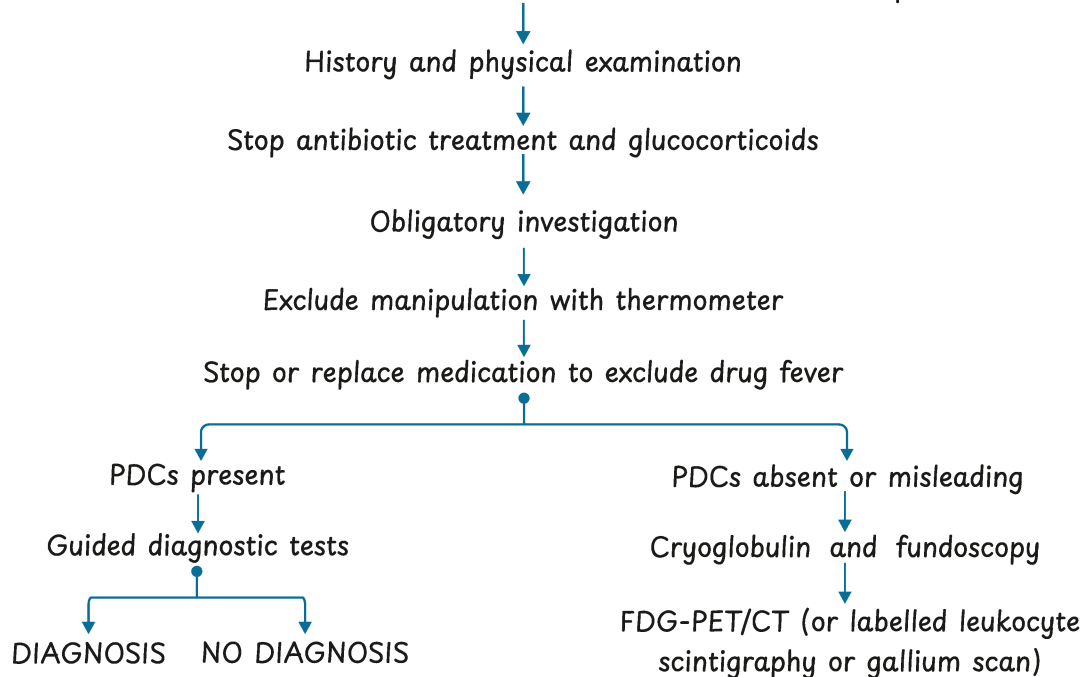
00:30:55

(Mnemonic: MMC)

- Exclude thermometer manipulation
- Stop/ replacing medications to identify drug fever
- Check for PDC (Potential diagnostic clues)
 - Headache: Lumbar puncture should be done → Send CSF for CBNAAT (To rule out Mycobacterium tuberculosis)
 - Cytopenia: Do Bone marrow biopsy → Send Culture (To rule out kala-azar)
- If PDC are absent
 - Check Cryoglobulins
 - Fundoscopy: Identifies
 - Roth spots: Seen in Infective endocarditis, Leukemias and Lymphomas
 - Choroid tubercles: Seen in Disseminated TB
 - FDG PET/CT: Takayasu arteritis (M/C blood vessel involved - left Subclavian artery) Takayasu arteritis (FDG-PET Scan)
 - Leucocyte scintigraphy: Radioisotope used - Indium-111
 - Chest CT/Abdominal CT - To rule out lymph node enlargement
 - Temporal artery biopsy - To rule out Giant cell arteritis, presents with temporal headache (Common cause of FUO in geriatric populations)

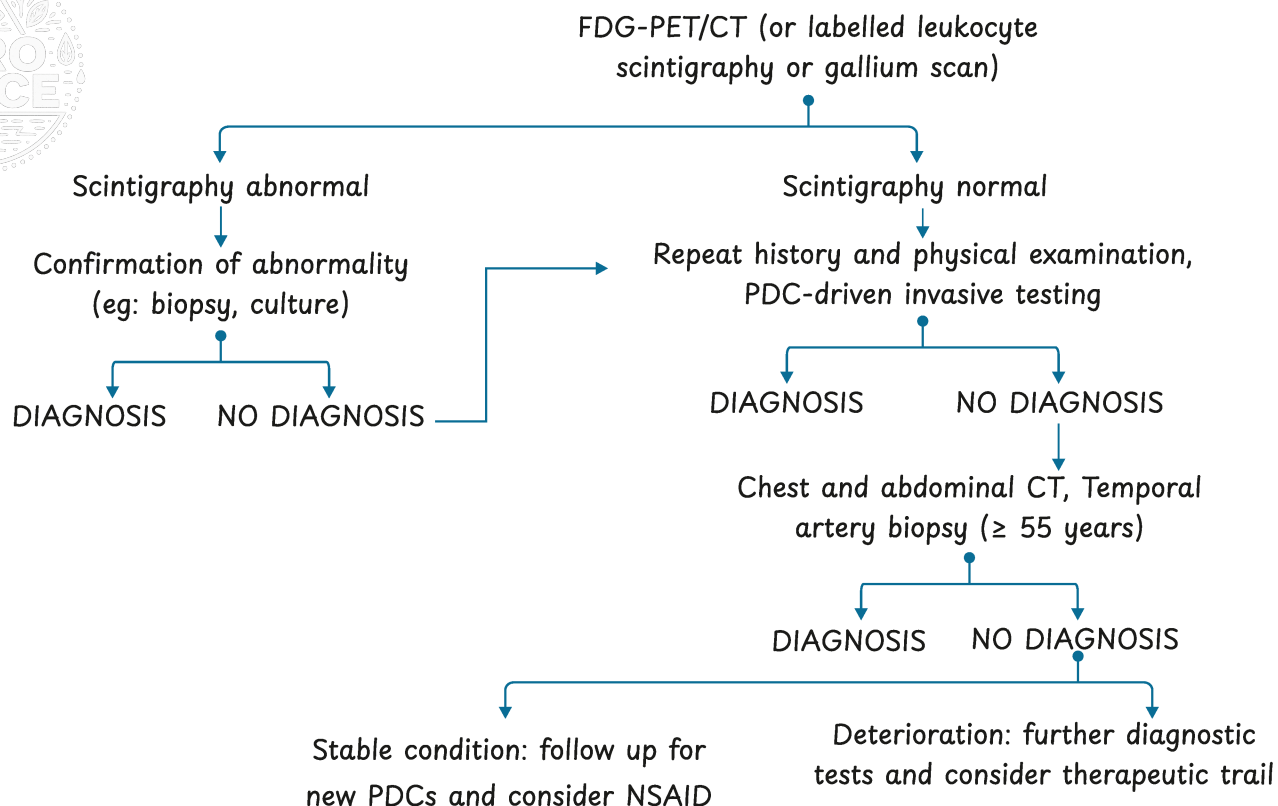


Fever $\geq 38.3^{\circ}\text{C}$ (101°F) and illness ≥ 3 weeks and no known immunocompromised state



Important Information

- Obligatory investigation: ESR or CRP, hemoglobin, platelet count, leukocyte count and differential count, electrolytes, creatinine, total protein, protein electrophoresis, alkaline phosphatase, AST, ALT, LDH, creatine kinase, antinuclear antibodies, rheumatoid factor, urine analysis, blood cultures (n=3), urine culture, chest X-ray, abdominal ultrasonography and tuberculin skin test



CAUSES

00:35:32

INFECTIOUS CAUSES	• TB • Endocarditis: Caused by <i>Coxiella burnetii</i> or <i>Bartonella Henselae</i> • Diverticulitis • Vertebral osteomyelitis • Q fever • Whipple disease: Caused by <i>Tropheryma whipplei</i> (cause Aortic stenosis) • Travel history-Malaria, kala-azar, Histoplasmosis (Cave disease) and Coccidioidomycosis (Valley fever)
NON- INFECTIOUS INFLAMMATORY DISEASES	• Large vessel vasculitis (Takayasu arteritis) • Sarcoidosis • Adult-onset Still's disease • Polymyalgia rheumatica
TUMOR	• Malignant lymphoma/leukemia
DRUGS	• Allopurinol • Antiepileptic drugs: Lamotrigine, Phenytoin • Sulfa drugs • Furosemide • Quinidine
FACTITIOUS FEVER	• Thermometer manipulation

- Fraudulent fever: Self-induced fever (By injecting distilled water or contaminated distilled water)



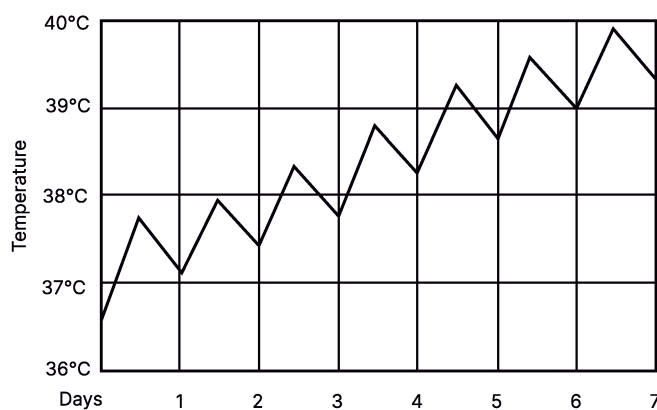
- Specific treatment
- Empirical Anti- tubercular treatment
- NSAIDs
- Colchicine
- Anakinra (IL-1 antagonist)



Q. Identify the pattern of fever being demonstrated in the graph.

- a. Sustained fever
- b. Remittent fever
- c. Intermittent fever
- d. Undulant fever

Ans (b)



Q. A 50year old male cardiac patient complaints of a fever of 38° C for the past 1 month. He denied any history of travel, does not have pets at home. Obligatory investigations showed an increased CRP only. What should be the next step in management?

- a. FDG PET
- b. Exclude manipulation of thermometer
- c. Leukocyte scintigraphy
- d. Stop all medications

Ans (b)



DENGUE FEVER



INTRODUCTION

00:03:35

- M/c Arbovirus infection
- Mortality rates
 - Dengue fever: 1%
 - Severe dengue: 5%
 - Undiagnosed dengue: 20% (D/t complications like Vasculopathy and Coagulopathy)
- Incubation period: 5-7days

WARNING SIGNS

00:04:37

(Mnemonic: BASIC-HP)

- **B**leeding: From gums, nose, GI track or mucosal sites
- **A**bdominal pain (persistent) along with nausea and protracted vomiting
 - Poor oral intake → Chances of patient deteriorating in circulatory failure increases
- **S**hock features like thready pulse, restless, obtundation (d/t ↓ in CNS perfusion)
- **I**rritability
- **C**linical fluid accumulation like Pleural effusion, hydrothorax and Ascites
- **H**epatomegaly
- **H**ematocrit ↑ by > 20% baseline admission value
- **P**latelet ↓

CLINICAL DIAGNOSIS OF DENGUE FEVER

00:06:54

- Fever in a person living in endemic zone with any 2 out of the 6 following features:
 - Retro-orbital pain, myalgia
 - Nausea and vomiting
 - **Rash** (present both in febrile and convalescence phase)
 - Warning signs
 - **Positive Tourniquet test** (According to CDC Government guidelines)
 - > 10 **Petechiae**/ square inch of Cubital fossa
 - TLC ↓

PHASES OF DENGUE

00:08:43

Febrile phase

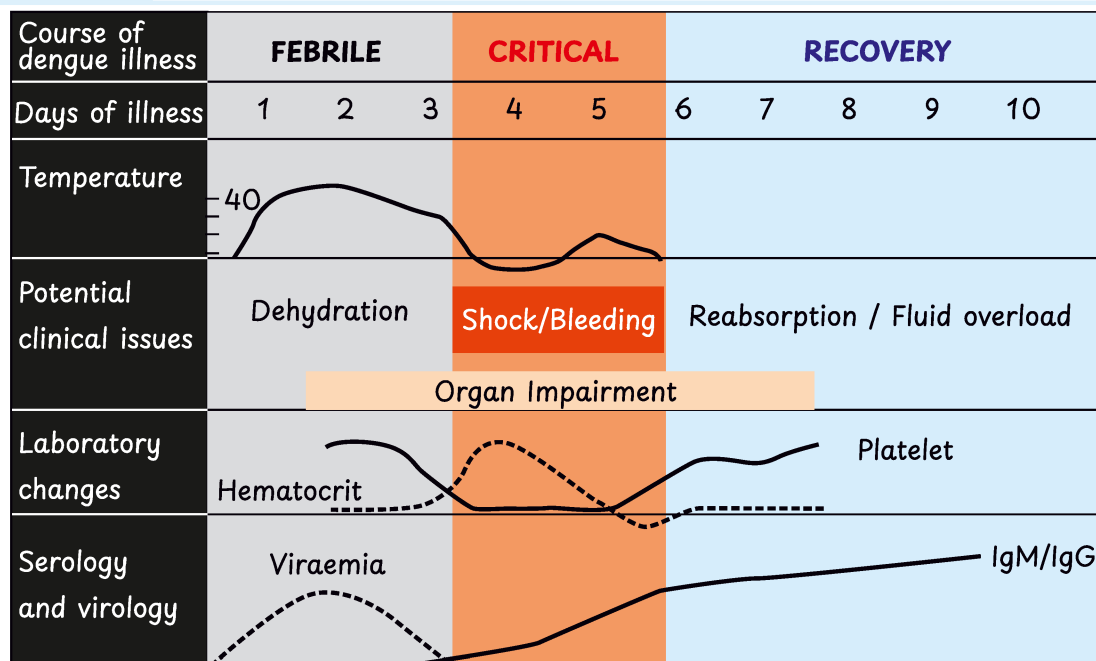
- Lasts 2-7 days
- Biphasic fever
- Clinical presentations
 - Headache
 - Cutaneous flushing
 - Vomiting
 - Myalgia and arthralgia
 - Maculopapular rash, appears on day 3-4 over chest, abdomen and face
 - Fades away as fever progresses



- 5-10% of the patients may progress to the critical phase
 - H/o previous dengue infection (secondary infection)
 - Occur after primary infection in patients with comorbidities and active coinfections.
- Begins after 3rd or 4th day of fever and may last about 24 - 48hrs
- Maintain hemodynamic stability of patient
- Ensure adequate platelets amount present in body
- Characterised by vasculopathy and coagulopathy → Plasma leakage → Excessive haemoconcentration → Bleeding → Shock and organ failure
- If dengue is not identified → patients die in undiagnosed fever

Convalescent phase

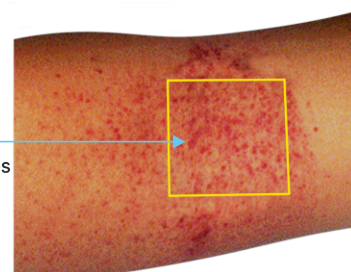
- Fluid returns to intravascular space
- Non judicious use of fluids → pulmonary edema
- **Confluent pruritic erythematous rash** with characteristic blanching is seen



TOURNIQUET TEST

- BP cuff is inflated between systolic and diastolic pressure
- Keep the cuff inflated for **2 minutes**
- > 10 petechiae per square inch of cubital fossa indicating capillary fragility (Positive tourniquet test)

>20 petechiae in a one-inch square indicates capillary fragility

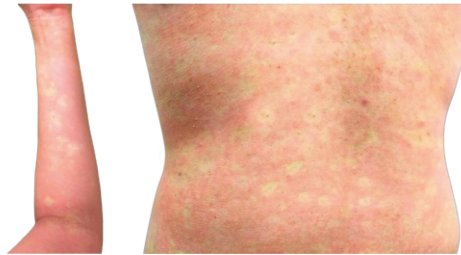


Important Information

- Trousseau Sign (Seen in **Hypocalcaemia/ tetany**)
 - Inflation of the BP cuff above the 20 mmHg of SBP
 - Wait for 3 minutes before sign appears
 - Carpopedal spasm



Trousseau Sign

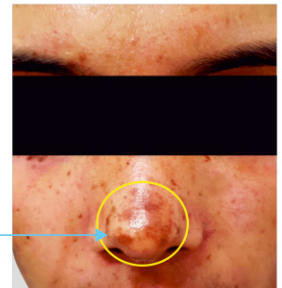


Dengue rash

Important Information

- History of fever and joint pain (arthritis) and after recovery patient develops hyperpigmentation on the nose or some other focal parts of body indicates a diagnosis of **Chikungunya**

H/o fever, joint pains with clinical resolution of the **hyperpigmentation** indicates a diagnosis of Chikungunya
Melasma



SEVERE DENGUE

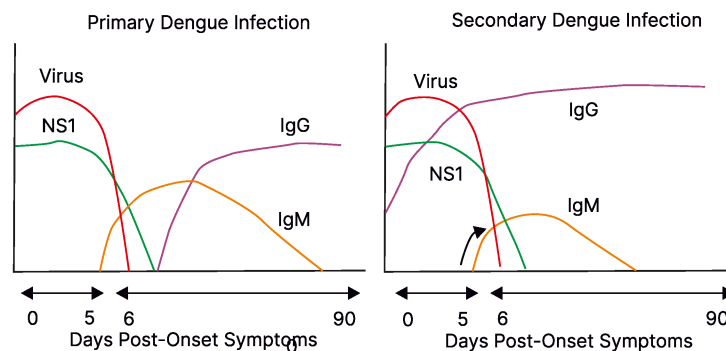
00:16.50

- Fever with any 1 of the following features (**Pneumonic: BPO**)
 - **B**leeding: Petechiae, mucosal bleeding
 - **P**lasma leakage/Fluid loss: Pleural effusion, Ascites, Hypotensive shock (Decompensated shock)
 - **O**rgan malfunction: Obtundation (Hypoperfusion of brain), elevated liver enzymes

Work Up

00:18:12

- **NS-1 Antigen** - Positive on **1- 4 days** of infection
- **MAC-ELISA** - Detects **IgM Antibody** → Positive on **4-5 days** of infection and continue to be present 3 months post symptoms and even persist longer
 - Detects antibodies against the enveloped proteins of the 4 serotypes of dengue (DENV 1-4)





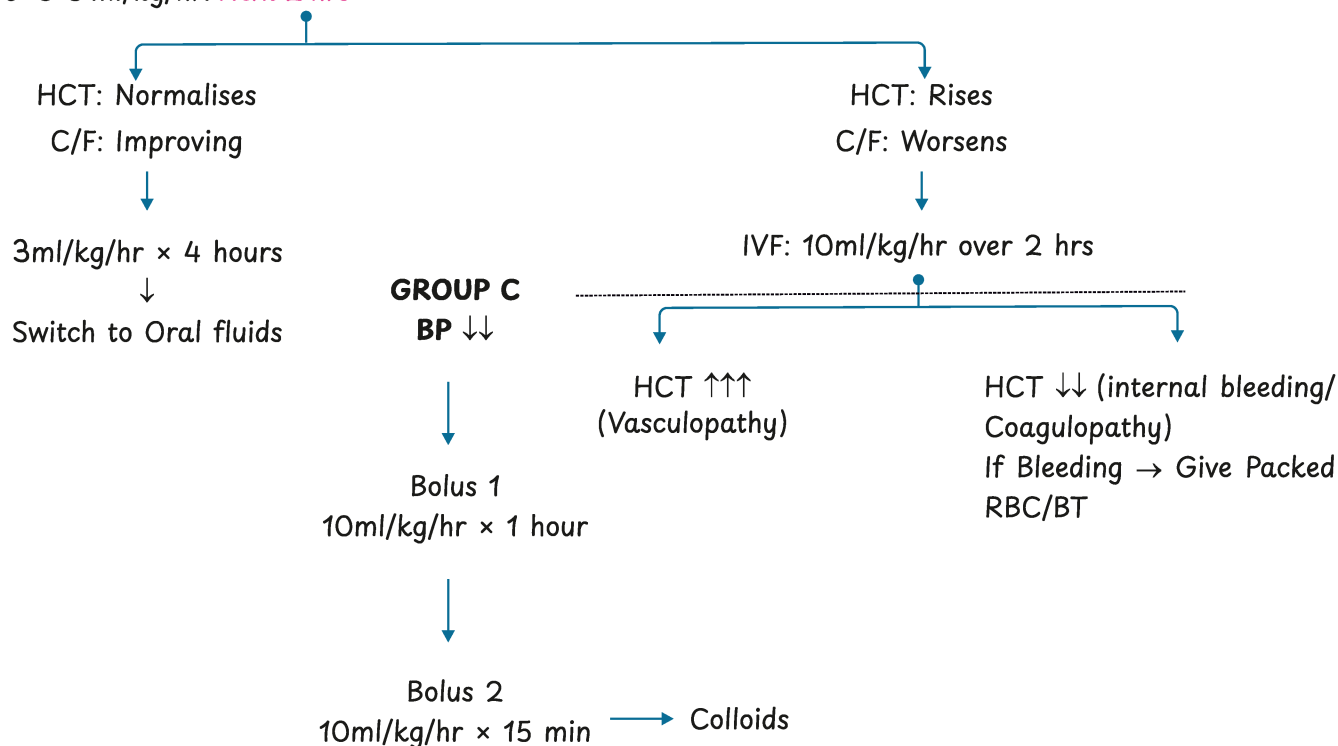
FEATURES	GROUP	INTERVENTION
Dengue without warning symptoms	A	OPD Basis
Dengue with warning symptoms or risk factors	B	Inpatient
Dengue with plasma leakage, bleeding, organ malfunction	C	Inpatient/ICU

MANAGEMENT OF GROUP A PATIENTS

- Awareness of critical period (From day 3/3.5 – 5/5.5 days)
- Adequate fluids to prevent dehydration
- Fever: Paracetamol
 - COX-1 inhibitors are contraindicated
- Insect repellent
- Watch **CCTV-R**
 - **C**olour, **C**apillary refill time, **T**emperature, **V**olume and **R**ate of pulse
 - If thready pulse → aggressive intervention

MANAGEMENT OF GROUP B PATIENTS

- Vitals monitoring
- I-O monitoring- Crystalloids: NS/RL
- Total – 10 ml/kg/hr over 4 hours
 - 5-7 ml/kg/hr: **First 2 hrs**
 - 3-5 ml/kg/hr: **Next 2 hrs**



DON'Ts IN TREATING A DENGUE PATIENT

00.26.10

- **Don't use corticosteroids**
- **Don't give platelet transfusions for a low platelet count/** mild gingival bleeding
 - If platelet count < 10,000 with severe bleeding - platelet transfusion can be given
- **Don't give half normal (0.45%) saline:** Worsens third space status
- **Don't assume that IV fluids are necessary:** Only 5-10% patients go into critical phase or show deterioration due to primary or secondary infection.

INDICATIONS FOR PLATELET TRANSFUSION IN DENGUE

00.26.49

Platelet count: < 10,000uL	• Indicated in the absence of bleeding, as the risk of spontaneous haemorrhage is high
Platelet count: 10-000 – 20,000uL	• Active mucosal or internal bleeding • Hemodynamic instability (↓ BP) • Planned invasive procedures
Platelet count: < 50,000uL	• Significant active bleeding, such as G.I hemorrhage or intracranial bleeding

- Patient with Petechial spots or mild mucosal bleed but **hemodynamically stable** → **supportive care** including bed rest, maintenance of hydration and monitoring.
- IM injections should be avoided.

Component	Volume(mL)	Content	Clinical response
PRBC	250-300 (including additive solution)	RBCs with variable leukocyte content and small amount of plasma	Increase haemoglobin 10g/L and hematocrit 3%
Platelets (from whole blood)	50-70 / RD unit pool of 4 to 6 RD unit	5.5×10^{10} /RD unit	Increase platelet count 5000-10,000 /μL/ RD unit
Platelets (from apheresis)	200-400	$\geq 2 \times 10^{11}$ /SDAP product	For pooled RD and SDAP: CCI $\geq 10 \times 10^9$/L within 1 h and $\geq 7.5 \times 10^9$/L within 24 hrs post transfusion
FFP	200-250	Plasma proteins coagulation	Increases coagulation factors by about 2%

DISCHARGE CRITERIA

00.29.58

- Patients who are resuscitated from shock rapidly recover.
- Patients with dengue haemorrhagic fever or dengue shock syndrome may be discharged from the hospital when they meet the following criteria.
 - Afebrile for 24 hours without antipyretics
 - Good appetite, clinically improved condition
 - Adequate urine output
 - Stable haematocrit level
 - At least 48 hours since recovery from shock
 - No respiratory distress
 - Platelet count greater than $> 50,000$ cells/ μ L

VACCINE FOR DENGUE

00.30.51

- Dengvaxia
 - CYD-TDV - tetravalent, live attenuated, chimeric dengue vaccine in a yellow fever 17D backbone developed by Sanofi Pasteur
 - Schedule - 3 doses of vaccine (At 0, 6 and 12 months)
 - Should be used in patients with past dengue virus infection



MCQ

- Q. Patient is admitted with diagnosis of dengue shock syndrome and is managed aggressively with fluids. Which of the following is not a criteria for discharge of this patient?
- a. 24 hours since recovery from shock
 - b. Haematocrit normalises
 - c. No evidence of breathing difficulty
 - d. Platelet count $> 50,000/\text{mm}^3$

Ans (a)



NIPAH, ZIKA AND MONKEYPOX



NIPAH VIRUS

00:01:50

- Initial cases were reported in 1999 in Malaysia in pig farm workers
- **In India**, the outbreak in Kerala is primarily d/t contact with either fruits contaminated with either saliva or fecal droppings of fruit bat
- **Family:** Paramyxovirus, ss negative sense RNA virus
- **Genus:** Henipa virus or Hendra virus (HeV)
- Naturally occurs in : **Bats/ pigs/ horses/ dogs**
- Death rate in outbreaks: 40-70%
- Cause of death : **Encephalitis/ ARDS**
- **Modes of Transmission:**
 - Consumption of fruits and date palm sap contaminated with saliva, fecal droppings of Pteropus
 - Man → man via body fluids
 - Pigs → man via body fluids



Pteropus (Fruit bat)

LIFE CYCLE

00:07:14

